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MILITARY HANDBOOK

AIRCRAFT FASTENER INSTALLATION AND REMOVAL METHODS FOR ATTACHED RETENTION ELEMENT (A.R.E.) SELF-LOCKING FLOATING NUT ASSEMBLIES

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MIL-HDBK-271-103
Aircraft Fastener Installation and Removal Methods for Attached Retention Element (A.R.E.) Self-Locking, Floating Nut Assemblies

1. This standardization handbook was developed by the Department of Defense in accordance with established procedure.

2. The publication was approved on 14 January 1986 for printing and inclusion in the military standardization handbook series.

3. This handbook provides installation and removal methods for attached retention element (A.R.E.) self-locking, floating nut assemblies. It will provide valuable information and guidance to aircraft mechanics at Army, Navy and Air Force Rework Facilities.

This handbook is not intended to be referenced in procurement specifications except for information purposes, nor shall it supersede any specification requirements.

4. Every effort has been made to include the latest information on installation and removal methods of A.R.E. nuts. Mechanics should review this handbook periodically. Users of this handbook are encouraged to report any errors or recommendations to the Commanding Officer, Naval Air Engineering Center, Code 93, Lakehurst, NJ 08733-5100.

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1. SCOPE

1.1 Scope. This handbook covers the installation, removal and repair methods for attached retention element (A.R.E.) floating nut assemblies.

1.2 Nut assembly types. There are two basic types of nut assemblies: the standard type and self-sealing type. The standard type is designed to accommodate various structural requirements, heavy shear loads, high corrosion steel requirements and oversize repair nuts. The self-sealing type is designed with a hermetic CRES steel housing and is provided with a elastomeric seal ring to prevent leak paths (see Figure 1).



FIGURE 1. Nut assembly types.

1.3 A.R.E. nut description. The standard nut assembly is made up of three components: a nut element, sleeve and cage (see Figure 2). The standard self-sealing nut assembly is made up of five components: a housing, nut element, sleeve, cage and seal (see Figure 3). The nut assembly is installed in a single hole by means of the sleeve component. The sleeve is inserted through a panel or workpiece and flared or swaged to captivate the nut to the structure (see Figure 4).

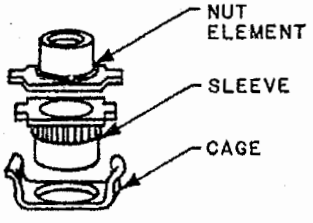


FIGURE 2. Standard nut components.

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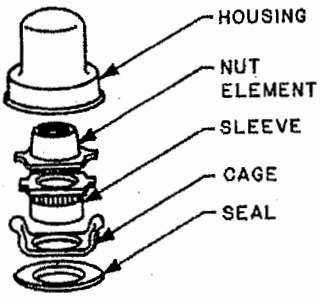


FIGURE 3. Seal nut components.

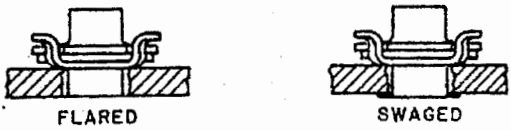


FIGURE 4. Mounting and installation.

1.4 A.R.E. nut applications. A.R.E. nuts are used in accordance with the design and usage limitations of MS14199 and MS33588. The nuts are designed to meet the requirements of MIL-N-25027.

2. REFERENCED DOCUMENTS

SPECIFICATIONS

MILITARY

MIL-N-25027	Nut, Self-Locking, 250°F, 450°F and 800°F

STANDARDS

MILITARY

MS14199	Nut Assembly, Floating, Self-Locking, Rivetless, Attached Retention Element (A.R.E.), Maintainability and Reliability Design and Usage Limitations for
MS14200	Nut, Floating, Self-Locking, Attached Retention Element (A.R.E.), 450°F, Steel, 125 KSI Ftu, Replaceable Nut Element

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STANDARDS (Continued)

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MS14201	Nut, Self-Floating, Self-Locking, Attached Retention Element (A.R.E.) 450°F, Steel, 125 KSI Ftu, Replaceable Nut Element, 1/64 Oversize Repair
MS14210	Nut, Floating, Self Locking, Attached Retention Element (A.R.E.) 450°F, Steel, 125 KSI Ftu, Replaceable Nut Element
MS14211	Nut, Floating, Self-Locking, Attached Retention Element (A.R.E.), 450°F, Cres Steel, 125 KSI Ftu, Replaceable Nut Element, 1/64 Oversize Repair
MS21051	Nut, Self-Locking, Plate, One Lug, Low Height, Alloy Steel, 125 KSI Ftu, 450°F
MS21060	Nut, Self-Locking, Plate, Two Lug, Floating, Low Height, Cres, 125 KSI Ftu, 450°F and 800°F
MS21061	Nut, Self-Locking, Plate, One Lug, Floating, Low Height, Alloy Steel, 125 KSI Ftu, 450°F
MS21062	Nut, self-Locking, Plate, One Lug, Floating, Low Height, Cres, 125 KSI Ftu, 450°F and 800°F
MS21076	Nut, Self-Locking, Plate, Two Lug, Floating, Reduced Rivet Spacing, Low Height, Cres, 125 KSI Ftu, 450°F and 800°F
MS33588	Nut, Self Locking, Aircraft Design and Usage Limitations of

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NATIONAL AEROSPACE STANDARDS (NAS)

NAS 1031	Nut, Self-Locking, Plate, Two Lug, Regular Height, Floating
NAS 1032	Nut, Self-Locking, Plate, One Lug, Regular Height, Non-Csk, Floating

(Application for copies should be addressed to National Standards Association, 5161 River Road, Washington, DC 20016.)

3. TOOLS AND GAGES

Use the following tools and gages to install and check the nut assemblies:

a. Drill bits
b. Installation tools
c. Gages

3.1 Drill bits. Use drill bits to drill the hole and simultaneously counterbore and countersink according to with the type of material to be drilled and the type of installation required (see 4.2.3).

3.2 Installation tools. Installation tools are available in four (4) groups.

a. Hand tools
b. Arbor press tools
c. Pneumatic tools
d. Air-hydraulic tools

Table I reflects some of the important features to consider when selecting an installation tool.

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TABLE I. Installation tool features.

Tool Capabilities	Hand Tool	Arbor Press Tool	Pneumatic Tool	Air Hydraulic
Small quantity installation	X	X	X	X
Large quantity installation		X	X	X
Single operation function		X		
Two operation function	X			
Swage/100° flare captivation	X	X	X	X
Nut thread engaged	X			
Nut thread not engaged		X		
High speed operation		X	X	X
High insertion for stainless steel				X
Portability	X		X	
Close Edge Dist. capability	X		X	X
Edge distance limited		X		
Normal hand force	X			
Finger pressure			X	X
Use for self sealing nuts				X
Three button operation				X

1/ Holding tools of the magnetic, collet or plier types are used to hold A.R.E. nut plates and self-sealing nut plates during installation with the hydraulic tool (see Figure 10).

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3.2.1 Hand tools. Use hand tools to install fasteners where the workpiece is too large to be brought under an arbor press and accessibility is a problem. Standard hand tools perform the installation by seating the lobes or by flaring or swaging the sleeves.

3.2.2 Arbor press tool. Use an arbor press tool to install fasteners where the workpiece can be placed under the ram, where edge distances are limited only by press throat depth and where the installation requires high insertion forces such as in stainless steel (see 4.1).

3.3 Pneumatic tool. A portable pneumatic tool requires an air supply of 90-110 psi (pound per square inch) and is a fast acting tool. Use this tool for a large number of installations at high speed and where edge distances are less than 1.400". Use the pneumatic tool for installations requiring high insertion forces such as in stainless steel panels (see 4.1).

3.4 Air-hydraulic tool. Use an air-hydraulic tool for a large number of installations where there is a high speed requirement and edge distance and accessibility are a problem. The air-hydraulic tool is considered to be the only power tool that works on self-sealing nut plates and requires the use of a holding tool to overcome the torque originated by the full thread mandrel going through the locking feature (see 4.1).

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4. INSTALLATION PROCEDURES

4.1 Selection of fasteners. Select the fastener in accordance with appropriate engineering criteria for the specific application. The basic features of the A.R.E. nut such as sleeve diameters and sleeve lengths should match the characteristics of the mounting structure. The panel thickness (p.t.) determines the fastener sleeve length (see Figures 5 and 6).

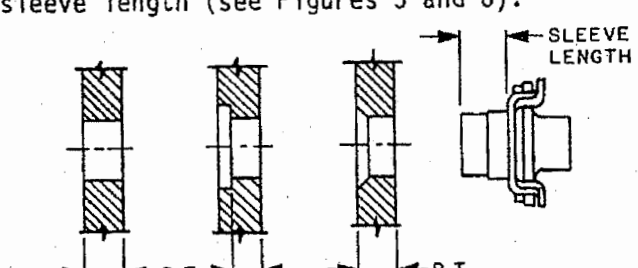
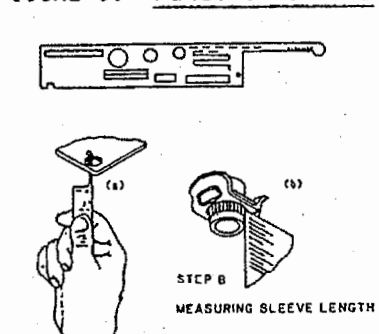


FIGURE 5. Panel thickness.



Note: (a) One side will measure the plain or counterbored panel thickness; the other will measure the countersunk panel thickness.

Note: (b) Code found on step "A" must match the code found on step "B."

FIGURE 6. Measuring panel thickness and sleeve length.

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4.2 Selection of tooling. Determine the selection of tooling by the type of installation structure material, type of fastener and accessibility (see Figures 7 thru 12). Installing fasteners in plain holes, counterbored holes and countersunk holes will require flaring tips for each set of tooling to properly install the fastener.

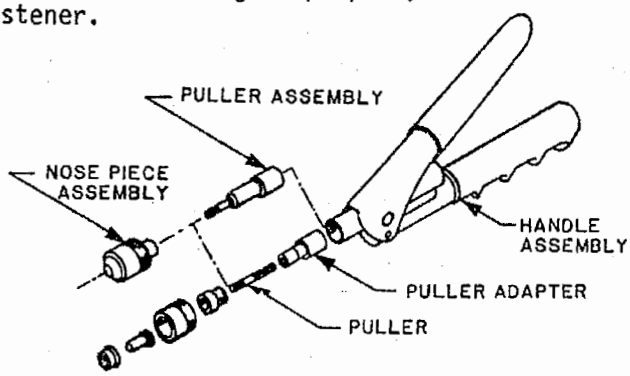


FIGURE 7. Standard hand tool for locking fasteners.

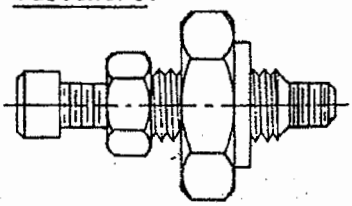


FIGURE 8. Hand tool for field installation.

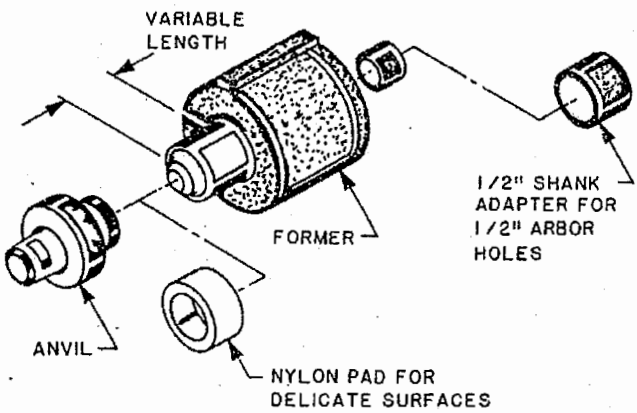


FIGURE 9. Arbor press tool.

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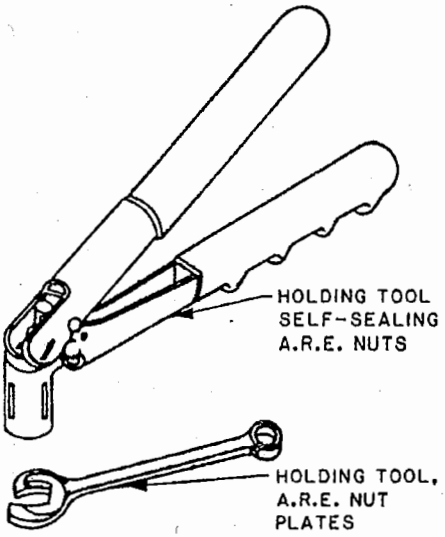


FIGURE 10. Holding tools.

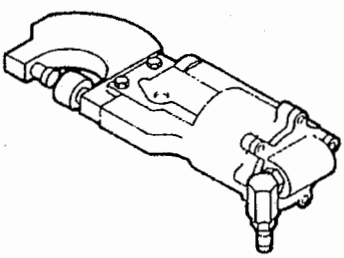


FIGURE 11. Pneumatic tool.

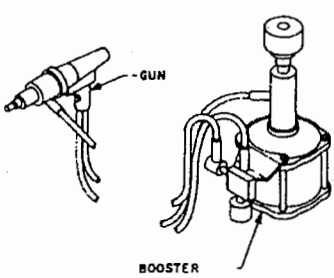


FIGURE 12. Air hydraulic tool.

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4.2.1 Tooling for types of material. A.R.E. nuts cannot be installed with a hand tool on material harder than HRB 93 unless the hole has been pre-broached. Special tools are required for panel thickness less than .045-inch (c'bored or plain hole) and .060-inch (c'sink). For non-metallics such as carbon-epoxy or Kevlar panels, special procedures and tools are required (see Table II).

TABLE II. Types of material.

Hardness	Hardness range 1/	Example of material
Soft	Less than BHN 100; HRB 62	Thermo plastics, fiber glass epoxy panels, unheat-treated aluminum alloys, brass, 5052 H32/H38
Hard	BHN 110-200; HRB 62-93	Heat treated aluminum alloys 2024 T3/T316 7075 T6/T73 5056 H18/H58
Hardset	Above BHN 200; over and including HRB 93	Inco 718/625 annealed, 431 Cres 4340 alloy, 300 series Cres, cold worked titanium 6AL-4V annealed only

1/ Definition of relative material hardness hand tools cannot be used on materials above HRB 93.

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4.2.2 Tooling for types of fasteners. Fasteners with standard diameter sleeves, reduced diameter sleeves, high shear sleeves and self-sealing nut plates require different tooling and tool adjustments to install each type of fastener.

4.2.3 Tooling accessibility. Fastener installations are either close to the edge or away from the edge. Within limits all tools will work on both applications. Hand tools do not have a limit; however, access to the backside is required for installation. The reach of the arbor press depends on the throat depth of the arbor press.

4.3 Hole preparation. Drill holes in accordance with Tables III and IV. Hole diameters shall be within prescribed tolerance limits including tolerance limitations on countersinks and counterbores. Prudent shop practice will determine the proper selection of combination drill bits for the required hole diameter, countersink or counterbore (see Figure 13).



FIGURE 13. Drill bit types.

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TABLE III. Standard and high shear hole diameters.

		Plain Hole	C'Bore Up To .075	Countersink	
	Size Dash No.	.030-.060	.051-Up	.055-.073	.074-Up
Standard diameter high shear sleeve	04, 06 08, 3 4	.217-.222 .276-.281 .343-.348	.219-.222 .280-.285 .347-.352	.217-.222 .276-.281 N/A	.219-.222 .280-.285 .247-.352

TABLE IV. Hole diameters for reduced diameter sleeves.

	Size Dash No.	Plain Hole
Reduced Diameter Sleeve	04	.1500-.1530
	06	.1750-.1780
	08	.2060-.2090
	3	.2345-.2375
	4	N/A

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4.4 Installation holes. Installation holes will vary when standard diameter sleeve A.R.E. nut plates, self-sealing nuts, high shear A.R.E. nuts and reduced diameter sleeve A.R.E. nuts are installed. Three types of installation are presently in use for the A.R.E. nut plate system (see Figure 14). A gasket with a punched hole is used when a recess is required to accommodate the swage in a plain hole installation. Countersinks and counterbores provide recesses for the swaged material so that two sheets of material are flush with each other.

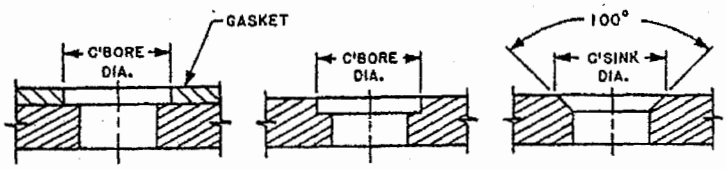


FIGURE 14. Types of installation.

4.4.1 Edge distance. Edge distances are controlled to meet stress requirements of the point and to prevent bulging due to press fit of the lobed sleeve. The edge distance can be computed by adding .090" to one half the diameter of the sleeve (see Figure 15).

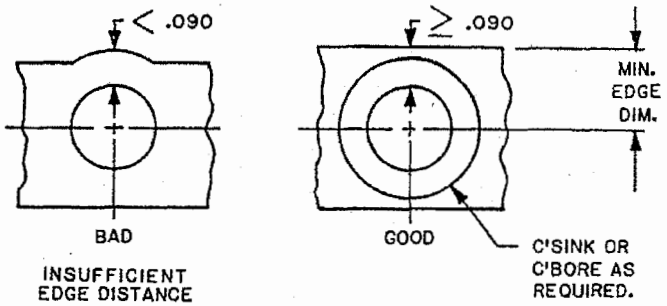


FIGURE 15. Edge distance.

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4.4.2 Hole quality and tolerance. Hole quality is the result of tools used. The A.R.E. nut plate will develop adequate torque on any type of hole within tolerance. Irregularity of holes, burrs and rifling will not degrade the hole performance. Tolerance control contributes to the performance of the A.R.E. nut but does not require particularly tight tolerances for good installations.

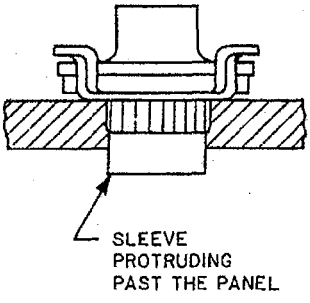


FIGURE 16. Lobes pulled into the panel.

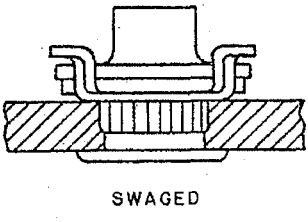


FIGURE 17. Fastener swaged or flanged against the panel.

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4.5 Fastening procedures. Install the A.R.E. nut plates by pulling the lobes into intimate contact with the panel (see Figure 16 and 17). A correctly adjusted hand tool will engage the threads but will not use the locking feature of the nut because of the relieved threads of the puller (see Figure 18).

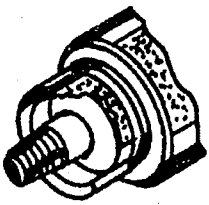


FIGURE 18. Relieved threads of puller.

4.6 Puller checks. Puller checks should be made to determine proper puller length, puller thread check to ensure correct thread to match the nut plate and to determine the correct puller for locking or non-locking nuts (see Figure 19).

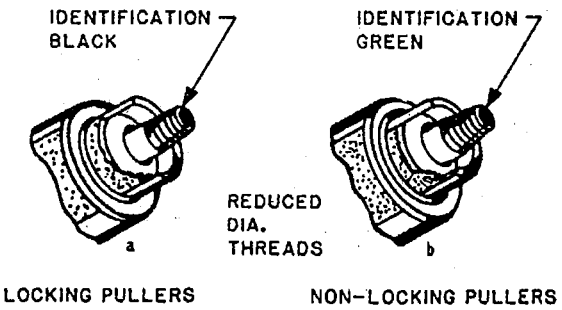


FIGURE 19. Nut plate puller.

4.6.1 Safety precaution. For safety, do not use locking (reduced) nuts because thread engagement may not be adequate for the required pull strength thereby endangering personnel with ejected nuts. The use of non-locking pullers on locking nuts is not recommended for the same reason.

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4.7 Fastener installation. Once the correct fastener has been selected, the appropriate hole drilled and the correct tool chosen, use the applicable tool to install the fastener. Established procedures in the sheet metal craft, together with engineering criteria, should be followed to successfully install correct plates.

4.7.1 Inspection of installed fasteners. Visually inspect inserted fasteners for lopsided and cracked upsets, periphery cracks, swaged protusions and cage seating.

4.7.1.1 Lopsided and cracked upsets. Lopsided and cracked upsets may not detract from the strength or performance of the fasteners but may signify a tooling problem (see Figure 20).

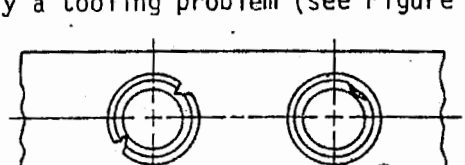


FIGURE 20. Tooling problem.

4.7.1.2 Periphery cracks. Cracks will not detract from torque or push-out performance of the nuts. A general acceptance criteria is that the flared or swaged section of the sleeve may have no more than one (1) medium crack. Severe cracks are unacceptable (see Figures 21, 22 and 23).

CRACKS ARE CLOSED AND ON THE PERIPHERY OF THE PART.

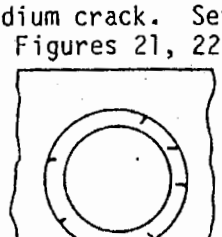


FIGURE 21. Light cracks.

CRACKS ARE OPEN AND EXTEND INWARDS ON THE SWAGE FLANGE.

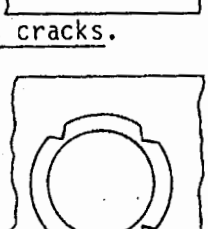


FIGURE 22. Medium cracks.

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CRACKS ARE OPEN AND EXTEND INTO THE CYLINDRICAL PORTION OF THE SLEEVE.

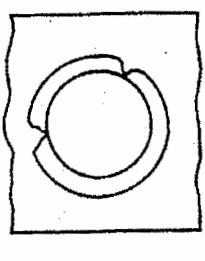


FIGURE 23. Severe cracks.

4.7.1.3 Swage protusions. Swaged protusions that are not flush with the surface of the material are unacceptable and should be rejected (see Figure 24).

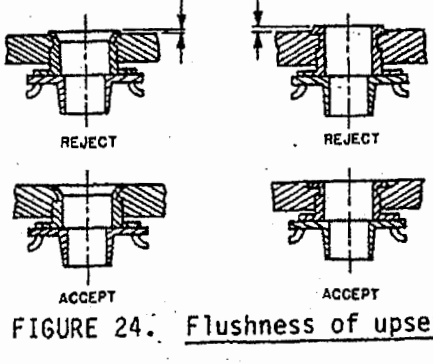


FIGURE 24. Flushness of upset.

4.7.1.4 Cage seating. The cage shall be seated flat against the panel so that a .002" shim cannot be freely inserted between the cage and the work piece. The cage is a sheet metal stamping and not always perfectly flat. The cage may not contact the panel all the way around; however, this condition is not detrimental provided the .002" shim is tight on a part of the cage (see Figures 25 and 26).

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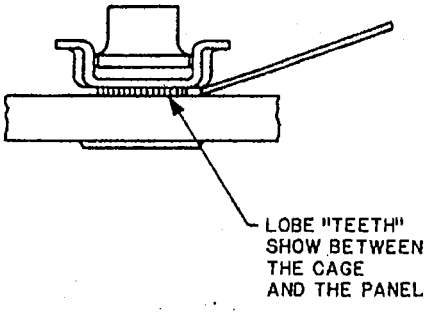


FIGURE 25. Unacceptable cage setting.

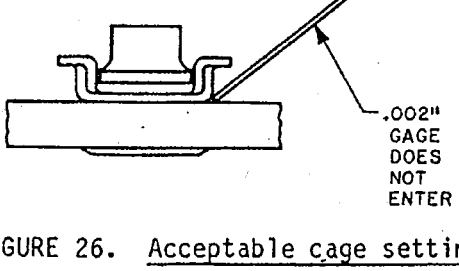


FIGURE 26. Acceptable cage setting.

4.7.1.5 Inspection of self-sealing nut. Self-sealing nut plates are installed to seal leak paths for gasses, vapors or liquids including aviation fields. Care should be taken to insure correct installation and should be inspected to determine proper gasket compression and nut height (see Figure 27, 28 and 29).

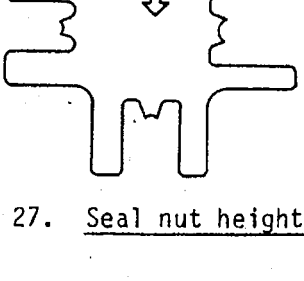


FIGURE 27. Seal nut height cage.

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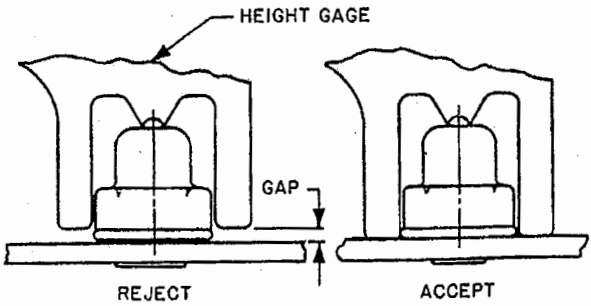


FIGURE 28. Installed nut height.

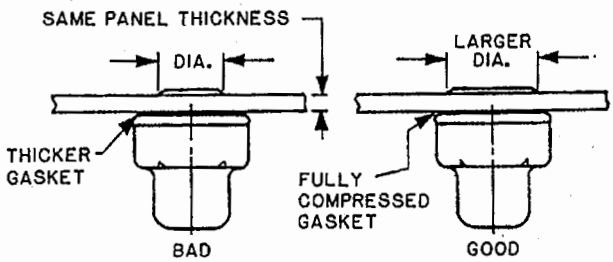


FIGURE 29. Gasket compression.

4.7.1.6 Pressure gage inspection of seal nuts. To test self-sealing A.R.E. nut plate installations for leaks, a screw-in type gage or a clamp-on gage is used to apply adequate air pressure to detect leaks.

4.7.1.6.1 Screw-in type gage. The screw-in type gage is usable in any location. This gage will measure leak rate by injecting a volume of air at any desired pressure up to 100 psi. A drop on the gage dial will signify a leak (see Figure 30).

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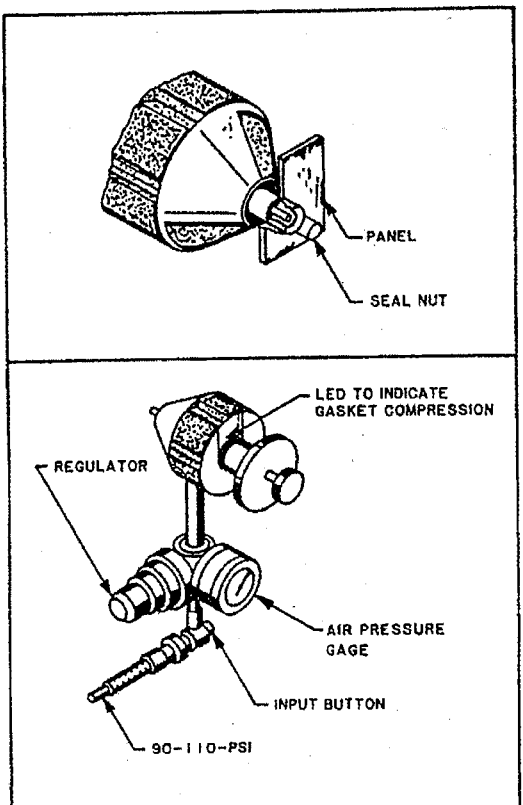


FIGURE 30. Gage, screw-in-type.

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4.7.1.6.2 Clamp-on type gage. The clamp-on type of gage is usable only where the parts to be tested are near the edge (1.00" maximum edge distance). A pressure shroud covers the part to be tested; the testing head moves forward to clamp the assembly from the possible side and adequate air pressure (100 psi max) is injected into the shroud. Any leakage will be detected by the transducer and a red fail indicator will turn on (see Figure 31).

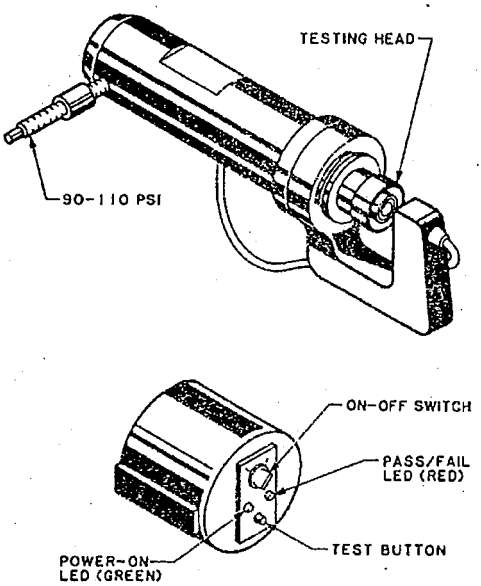


FIGURE 31. Gage (clamp on type).

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5. NUT REMOVAL AND ELEMENT REPLACEMENT PROCEDURES.

5.1 A.R.E. nut assembly removal. Remove A.R.E. nut assemblies when torque performance is inadequate or when the wrong nut assemblies are installed. Removal is also required due to severe cracks on swage or flare, broken cages or collapsed sleeves. Any assembly must be removed by machining the flare or swage.

5.1.1 Recommended removal method. Use an adjustable microstop tool to mill the swage or flare because adequate tool protrusion is obtained and there is less chance of tracking the panel (see Figure 32).

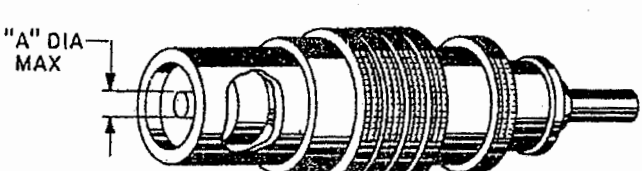


FIGURE 32. Microstop countersunk tool.

5.2 Removal tools.

5.3 A.R.E. nut element removal tool. Use a nut element removal tool to remove nut elements from installed assemblies. The removal tool is available in many configurations depending on the accessibility of the installed assembly (see Figure 33).

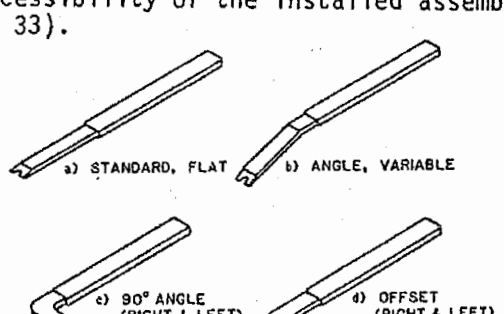


FIGURE 33. Types of removal tools.

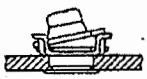
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5.4 A.R.E. nut assembly removal tool. Use the nut assembly removal tool to countersink flared or swage parts, weakening the flare, to remove an installed nut assembly (see Figure 32). Use the tool with a 2000 rpm (minimum) hand drill motor or drill press.

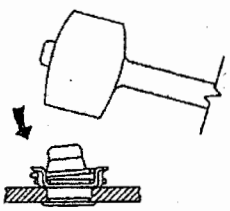
5.5 Replacement procedure.

5.5.1 A.R.E. nut element replacement. Replace A.R.E. nut elements when locking is no longer effective, thread damages are apparent or a change is required in material, thread designation, plating or lubrication.

Using the appropriate tool, remove the nut element by following instructions in Figure 34. Replace the nut element by following instructions in Figure 35.



STEP 1: PLACE NUT ELEMENT INTO CAGE ONE EAR FIRST.



STEP 2: STRIKE LIGHTLY TOP OF NUT WITH A SMALL HAMMER OR MALLET TO SNAP SECOND EAR INTO THE CAGE.

FIGURE 34. Removal nut element.

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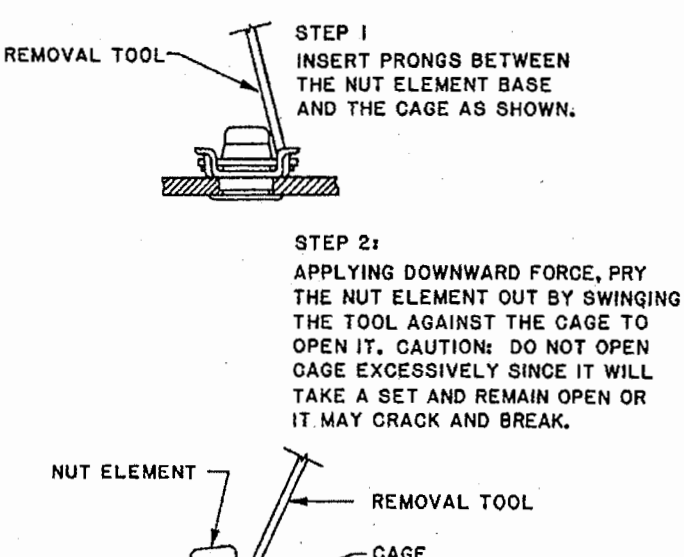


FIGURE 35. Installation nut element.

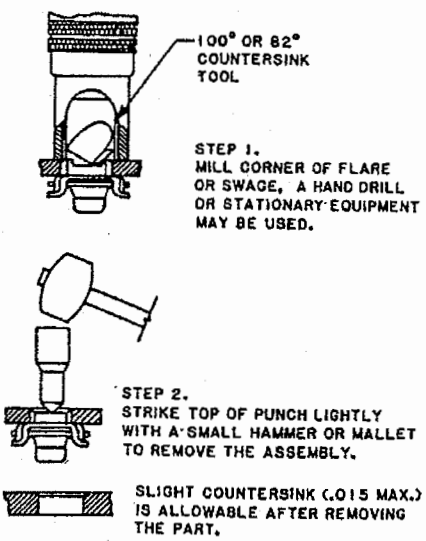


FIGURE 36. A.R.E. nut assembly removal.

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5.5.2 Optimal removal methods. The use of micro-stop countersink is always recommended to be used to reduce possible panel damage. However, the following optimal methods can be used with caution.

5.5.3 Drilling swage or flare. Under certain conditions drilling the swage or flare can be done; however, there is a risk of the sleeve turning and reaming the hole (see Figure 37).

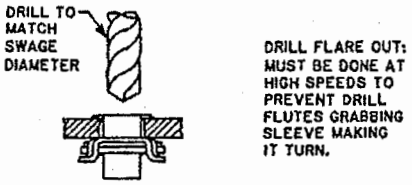


FIGURE 37. Drilling flare or swage.

5.5.3.1 Milling swage or flare. A preferred method is to use a countersink tool to mill the corner of the swage or flare (see Figure 38). An alternate method is to mill the entire flange exercising care not to touch the panel (see Figure 39).

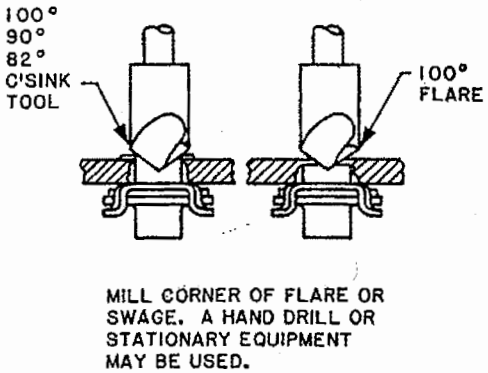


FIGURE 38. Milling flare or swage (preferred method).

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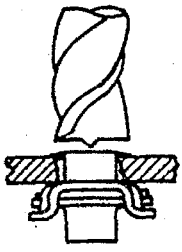


FIGURE 39. Milling flare or swage (alternate method).

5.5.4 Reaming swage or flare. Ream the flare or swage to remove the assembly by using a reamer with a diameter equivalent to the original hole diameter (see Figure 40 and Table V).

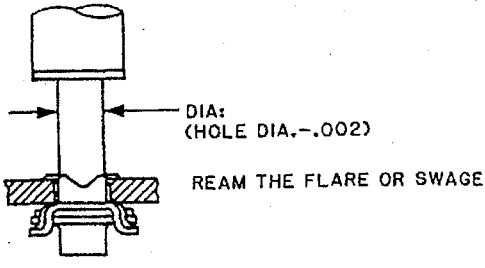


FIGURE 40. Reaming flare or swage.

TABLE V. Reamer diameters.

REAMER DIAMETERS FOR A.R.E. NUT REMOVAL		
Size Dash No.	Standard Diameter and High Shear Sleeves	Reduced Diameter Sleeves
04	.218-.216	.156-.154
06	.218-.216	.175-.173
08	.281-.279	.195-.193
3	.281-.279	.234-.232
4	.348-.346	.297-.295

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5.6 Damaged holes. While removing installed A.R.E. nut elements, the hole in the panel may become enlarged or damaged by torque.

5.7 Gages. Gages are classified in three (3) categories:

a. Panel thickness gage
b. A.R.E. self locking nut height
c. A.R.E. self sealing nut pressure tester

5.8 Panel thickness gage. Use the panel thickness gage for measuring panel thickness, checking nut heights and checking sleeve diameters (see 4.1).

5.8.1 Seal nut height gage. Use the seal nut height gage for measuring installed nut height variations and for measuring the nut height after installation (see 4.7.1.5).

5.8.2 Seal nut pressure gage. Use the seal nut pressure gage for testing self-sealing A.R.E. nut plate installations. Use the pressure gages to apply air pressure to detect leaks (see 4.7.1.6 and 4.7.1.6.1).

5.8.3 Rotation of nut assembly sleeve. If the nut assembly sleeve rotates in the hole under torque, use A.R.E. nuts that have been oversize hole diameters to engage material outside of the periphery of the damaged hole (see Table VI).

TABLE VI. Table of oversize nuts.

Repair Nut	Use to Repair
MS14201 MS14211	MS14200 MS14210

5.8.4 Enlarged holes. Standard riveted nut plates of the same characteristics of the affected A.R.E. nut may be installed. For application requiring a close-fit hole-screw shank tolerance, a grommet or bushing can be used (see Tables VII and VIII and Figure 41).

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TABLE VII. Table of riveted nut plates.

A.R.E. Nuts 1/64 Oversize Repair	Riveted Nut Plates
MS14201	NAS1031, MS21051, MS21060, NAS1032, MS21061, MS21062
MS14211	MS21060 (450°) MS21076 (450°)

TABLE VIII. Table of bushing diameters.

Size Dash No.	A Dia.
04	.156-.159
06	.175-.178
08	.195-.198
3	.234-.237
4	.297-.300

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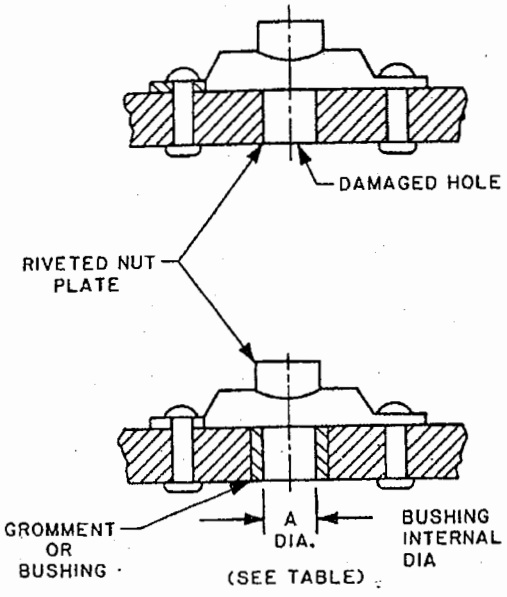


FIGURE 41. Nut plate installation.

5.9 Troubleshooting guide.

5.9.1 Flare cracks. The following condition can be the cause of flare cracks and should be corrected.

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Probable Cause	Action
a. Bad tooling	Check tool tip for tool marks, chipped end or wrong flaring tip, lubricate, polish or replace tool if found to be defective.
b. Tool misalignment	Check puller straightness or arbor press alignment, replace puller or adjust arbor press.
c. Excessive flare	Check parts length versus the panel thickness (per specification) replace parts with correct ones.
d. Countersink too wide	Check c'sink tool setting, adjust.
e. Knicks and imperfections on sleeve end.	Visually inspect the parts. Replace if necessary.
f. Sleeves made of incorrect material (sleeve cracks on all parts) material cold-worked (all parts crack severely).	Analysis of material. It should be 304 or 305 cres. Replace parts if tooling checks okay.

5.10 Torque-out below specification. All rivetless nut plates shall meet or exceed MIL-N-25027 requirements.

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Probable Cause	Action
a. Hole oversize	Remove nut assembly. Install repair nut. Check all holes.
b. Parent material too hard (lobes sheared)	Remove nut assembly. Check material hardness. Install new nut using proper tooling and method.
c. Parent material too soft (lobes rotated)	Remove nut assembly. Choose the correct tools and methods. Install repair nut on removed installation.
d. Hole too small (lobes collapsed or sheared)	Remove nut assembly. Check hole sizes against the recommended hole size. Open-up or redrill hole if required. Install new nut using proper tooling and method.
e. Improper nut seating (lobes show between the panel and cage)	Re-strike the nut with the appropriate tool until seated correctly. Check torque. If low, remove and install repair nut.
f. Panel buckled during flaring operation (on thin panels, not enough lobe enlargement is achieved)	Remove nut; flatten panel if possible. Check tool listing for proper tooling and method; install new nut, or if hole oversize, install a repair nut using proper methods.

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5.11 Nut element does not float before installation.

Probable Cause	Action
a. Cage bent (see Figure 42).	Install part; it will straighten up on installation. If nut still binds, close bracket ears slightly with pliers. If nut still binds, reject nut.

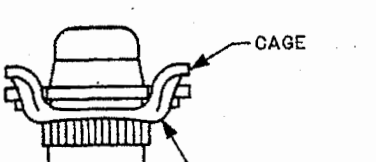


FIGURE 42. Bent cage.

b. Nut ears bent (most likely to happen on size 04 and 06) (see Figure 43).	Reject nuts.

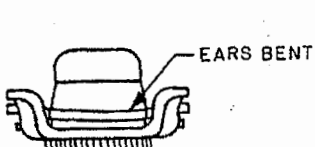


FIGURE 43. Ears bent.

c. Nut or sleeve ears too thick.	Reject nut.

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5.12 Nut does not accept mating part or gage.

Probable Cause	Action
a. Incorrect threads	Check for metric, quad lead threads, remove nut element and replace with correct nut.
b. Locking threads	Remove nut element and replace.
c. Excessive dri-lube or plating	Run cold forming tap thru each part.
d. Cross-threads	Visually inspect parts, return bad parts. Remove nut element and replace.
e. Burrs on thread	Visually inspect parts, run cold forming tap thru each part.

5.13 Nut element falls out of cage.

Probable Cause	Action
a. Cage too open.	Close bracket slightly with pliers.
b. Nut ears too small.	Close bracket slightly with pliers.

5.14 Seal nut leaks under pressure.

Probable Cause	Action
a. Nut pulled in all the way (gasket not fully compressed evidenced by higher nut profile using height gage or smaller swage than normal).	Re-strike nut using appropriate tooling to pull the nut in the lobes. This can be visually checked. Check air-hydraulic pressure, bleed if required.
b. Housing swage not wrapped on the seal completely (see Figure 44).	Remove nut and install new or add bead of sealant around the nut base on installing nut.

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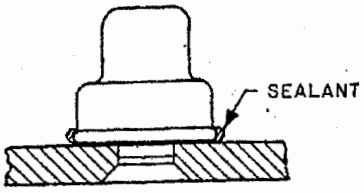


FIGURE 44. Incomplete seal.

c.	Panel surface not perpendicular to hole causing uneven seal compression (see Figure 45).	Spot face the panel or add a sealant foundation to the panel prior to installing the nut.

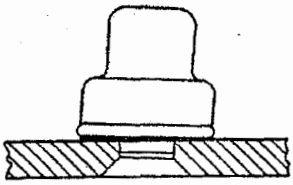


FIGURE 45. Uneven seal compression.

d.	Panel buckled during the installation precluding good seal contact (see Figure 46).	Check for proper method on installation. Remove nut; flatten panel if possible, add a bead of sealant around the hole, and install new nut.

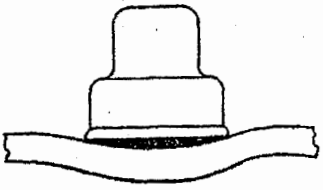


FIGURE 46. Buckled panel, bad seal contact.

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e.	Nut housing pushed back or punctured (see Figure 47).	Puller on tool too long; check puller. Remove nut and replace with a new nut.

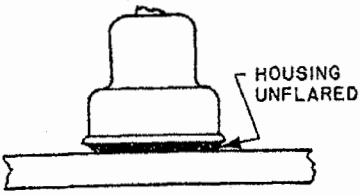


FIGURE 47. Punctured nut housing.

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Army - AV
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Preparing Activity:
Navy - AS
(Project No. 53GP-0149-06)

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